Classification from Partitioned Datasets with Test-Time Side Information

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Motivation – University Admissions

The Scenario

- A **primary university** predicts applicants' aptitude from profiles.
- At test time, an applicant also discloses a **secondary university's decision** (side information).
- Goal: Leverage this decision to improve the primary university's accuracy.

The Challenge

- No candidate in historical data has **both** admission outcomes simultaneously.
- Training data is **partitioned**: features + primary labels *or* features + secondary labels.
- The joint distribution (X, Y, Z) is unlearnable from partitioned data.

Problem Setup: Partitioned Dataset

Spaces: Feature $\mathcal{X} \subset \mathbb{R}^d$, Label $\mathcal{Y} = \{-1, 1\}$, Side Info $\mathcal{Z} = \{-1, 1\}$.

Partitioned Training Data

- Primary dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$ (missing z_i)
- Secondary dataset $S = \{(x_j, z_j)\}_{j=n+1}^{n+m}$ (missing y_j)

All samples drawn i.i.d. from the joint (X, Y, Z), but **labels and side info are never observed together**.

Goal: At test time, given (x, z), predict label $\hat{y}$ using a classifier learned from *both* datasets.

Key Definitions: Generative Models

We employ three logistic regression classifiers thresholded at 0.5:

1. **Side-Info Classifier**: Predicts Z from X.

$$\phi_u(x) = 2\mathbb{1}_{\{\sigma(u_c + u_x^\top x) > 0.5\}} - 1 \quad \rightarrow \quad \textbf{Optimal: } u^\star$$

2. **Primary Classifier**: Predicts Y from X alone.

$$\psi_v(x) = 2\mathbb{1}_{\{\sigma(v_c + v_x^\top x) > 0.5\}} - 1 \quad \rightarrow \quad \textbf{Optimal: } v^\star$$

3. **Composite Classifier**: Predicts Y from (X, Z).

$$\pi_w(x, z) = 2\mathbb{1}_{\{\sigma(w_c + w_x^\top x + w_z z) > 0.5\}} - 1 \quad \rightarrow \quad \textbf{Optimal: } w^\star$$

Algorithm: Learning with Side Information

Two-Stage Imputation Procedure:

Stage 1: Imputation Model

Learn $X \to Z$ mapping $\hat{u}$ by minimizing regularized empirical risk on dataset S. This defines the side-info classifier $\hat{\phi}$.

Stage 2: Composite Classifier

1. For each $(x_i, y_i) \in D$, **impute** $z_i = \hat{\phi}(x_i)$.
2. Train regularized composite classifier $\hat{w}_\lambda$ on D using imputed z_i.
3. Regularization: $\lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2$.

λ_z is the critical hyperparameter controlling trust in imputed Z.

Main Result: Upper Bound on Excess Risk

Theorem (Main Theorem)

With high probability, the excess population risk is bounded by:

$$
\begin{aligned}
R(\hat{w}_\lambda) - R(w^\star) \;\leq\; & \underbrace{\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)}_{\text{Estimation Error}} \\
& + \underbrace{\min\left\{\mathcal{O}(\lambda_z), \mathcal{O}(1)\right\} \cdot \left(T(v^\star) - R(w^\star)\right)}_{\text{Regularization Bias}} \\
& + \underbrace{\min\left\{\mathcal{O}\left(\frac{1}{\lambda_z}\right), \mathcal{O}(1)\right\} \cdot \left(\mathcal{O}\left(\frac{1}{\sqrt{m}}\right) + L(u^\star)\right)}_{\text{Imputation Error}}
\end{aligned}
$$

The λ_z Trade-Off

Small λ_z (Full Trust)

- Encourages use of side info Z.
- **Regularization Bias** $\downarrow$
- **Imputation Error** $\uparrow$
- Best when Z is informative and predictable from X.

Large λ_z (No Trust)

- Ignores side info Z in prediction.
- **Regularization Bias** $\uparrow$
- **Imputation Error** $\downarrow$
- Best when Z is noisy or poorly imputed.

Bang-Bang Optimality of Regularizer

Key Takeaway

For large n, m and $L(u^\star) \leq 1$, the optimal regularizer $\Lambda_z^\star$ is either ∞ or 0.

- **Regime I**: If side info is noisy/uninformative $\implies \Lambda_z^\star = \infty$ (completely ignore).
- **Regime II**: If side info is clean/informative $\implies \Lambda_z^\star = 0$ (fully trust).
- There is no middle ground! The model should never partially trust the imputed side information in the asymptotic limit.

Lower Bound: A Fundamental Limitation

Theorem (Lower Bound on Excess Risk)

For **any** learning algorithm A trained on partitioned dataset (D, S), there exists a valid joint distribution of (X, Y, Z) such that:

$$\mathbb{P}\left[R(A(D,S)) - R(w^\star) > 0.19\right] \geq \frac{1}{2}$$

Proof Idea:

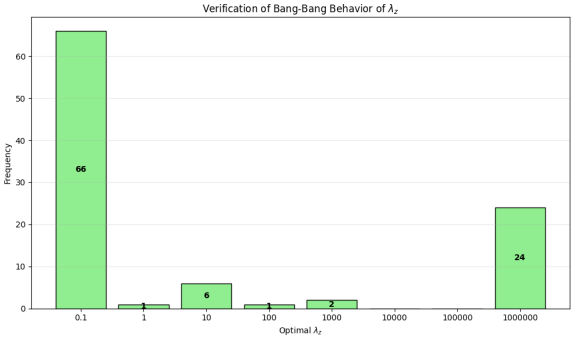
- Construct two joint distributions with **identical marginals** (X, Y) and (X, Z).
- Partitioned datasets only reveal marginals!
- No algorithm can distinguish the two distributions, yielding an irreducible error term regardless of infinite n, m.

Simulation Results

Setup:

- 100 random Gaussian mixture distributions for (X, Y, Z)
- $m = 1000$, $n = 4000$
- Candidate $\lambda_z \in \{10^{-2}, 10^{-1}, \ldots, 10^5\}$

Observation: Optimal λ_z aggressively concentrates at the extreme edges! This verifies the theoretical bang-bang behavior.



Conclusion

- **The Core Issue**: Classification using side information from partitioned datasets where features, labels, and side info are never jointly observed.
- **The Method**: A two-stage imputation algorithm utilizing a critical regularization hyperparameter λ_z.
- **The Math**: Explicit upper bounds on excess risk featuring a tension between estimation and imputation.
- **The Surprise**: The optimal choice of λ_z is *binary* — full trust or complete ignorance.
- **The Limit**: Even infinite datasets cannot close the gap due to unlearnable joint distributions.

Thank You!